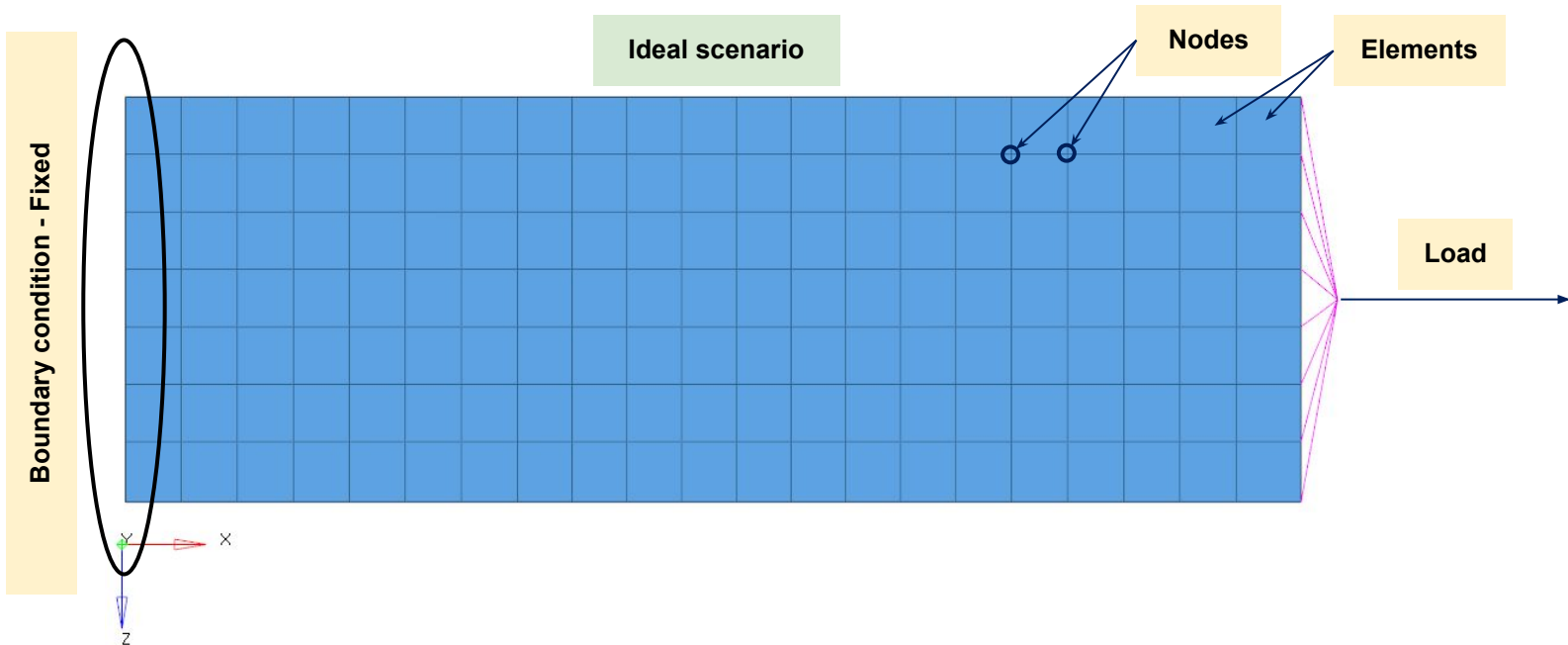
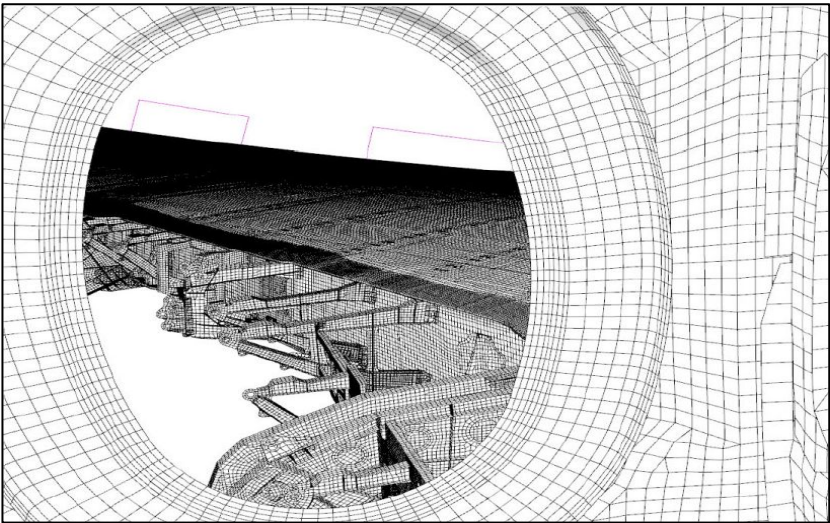


Idea explanation with examples

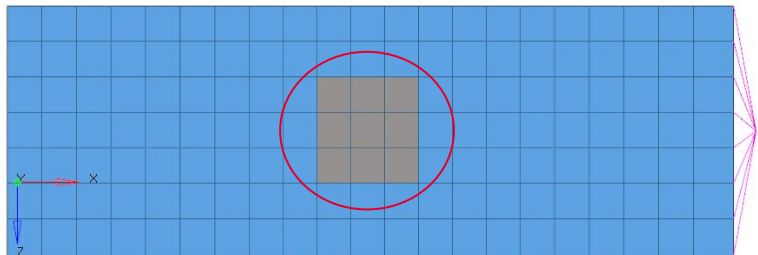
Swarnendu Samanta
Jaya Raju Namala
Anand Nagarajan

AIRBUS

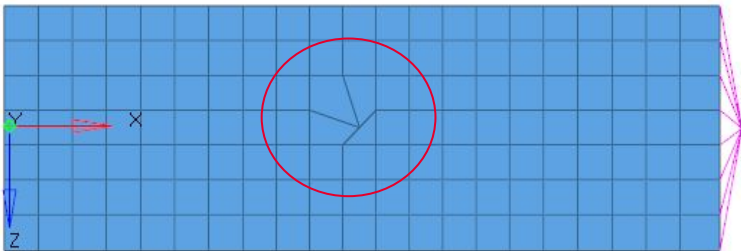
Typical Aircraft Finite Element Model



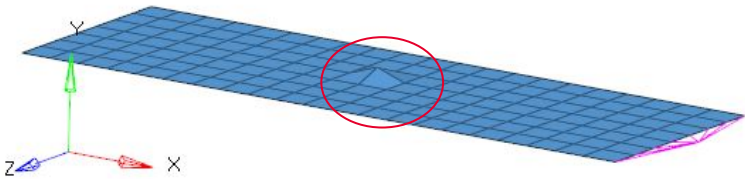
Case I: Duplicate elements



Case II: In-plane distortion



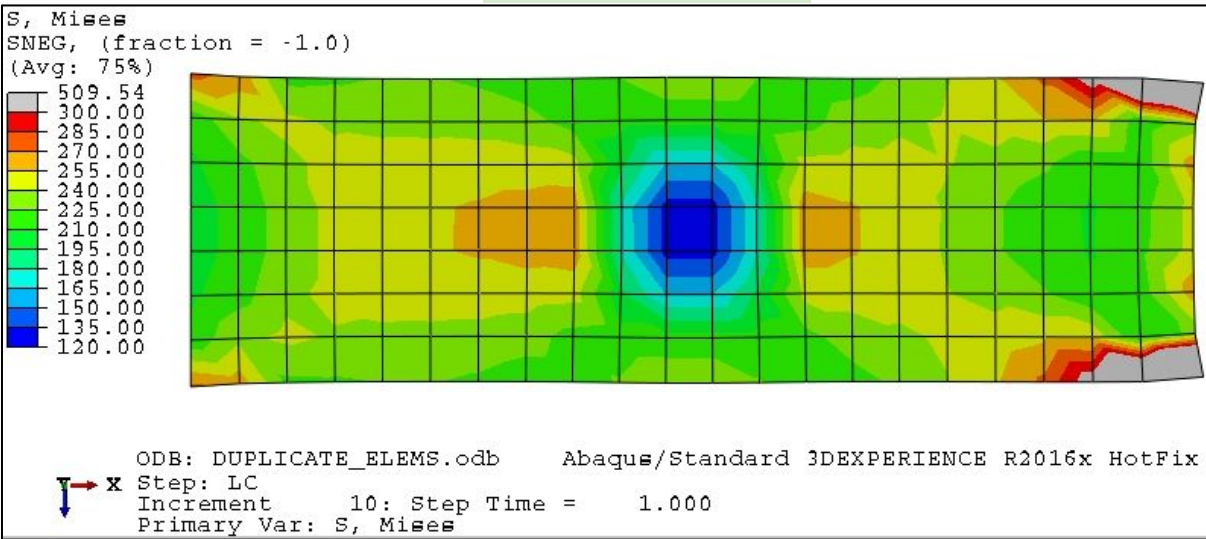
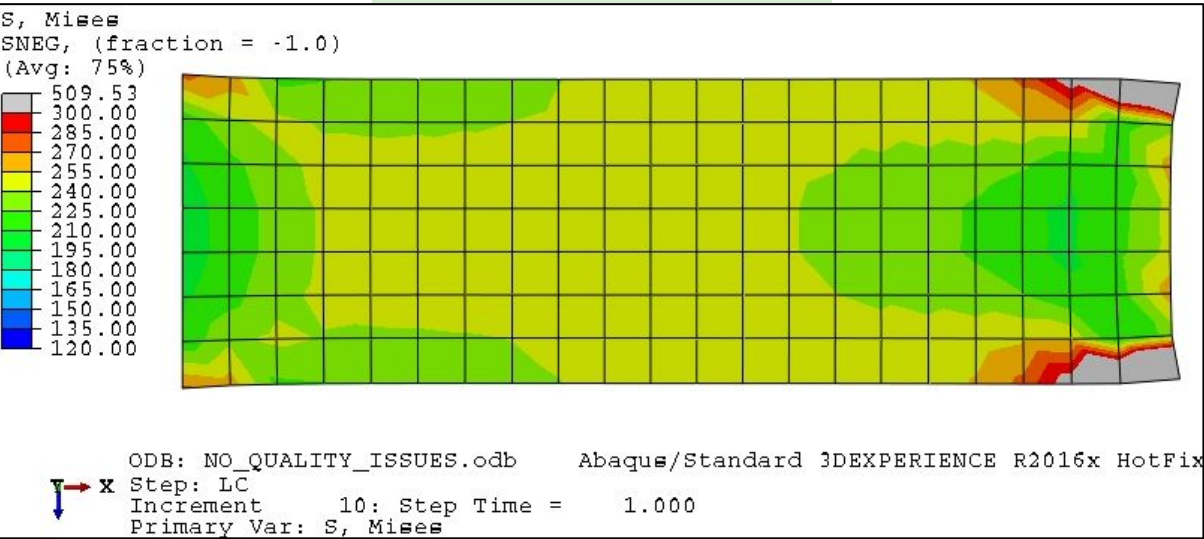
Case III: Out of plane distortion



Difference in stress values (different patterns)

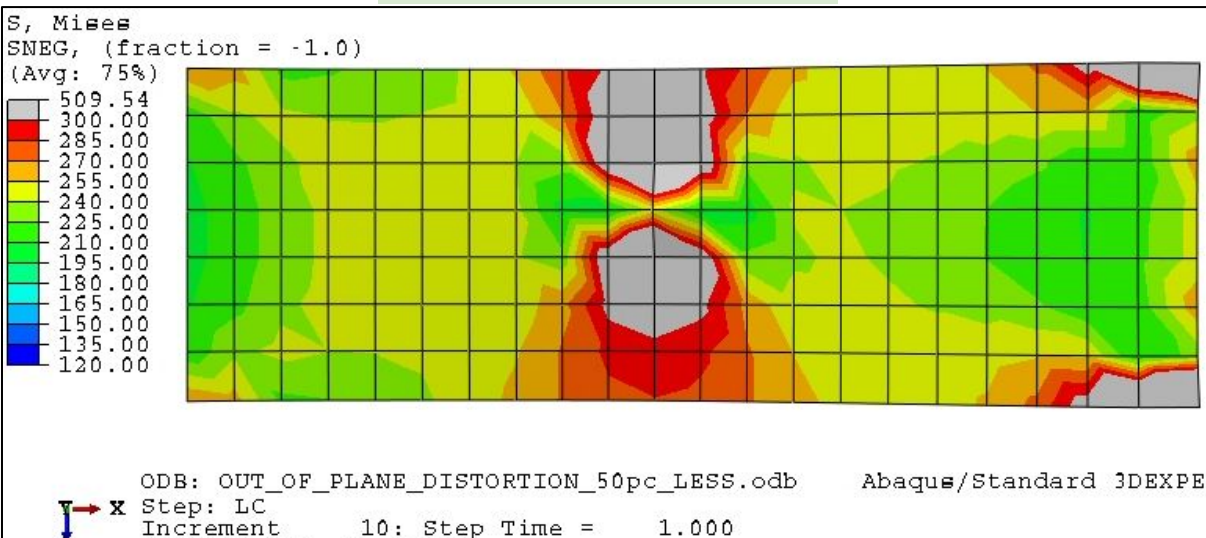
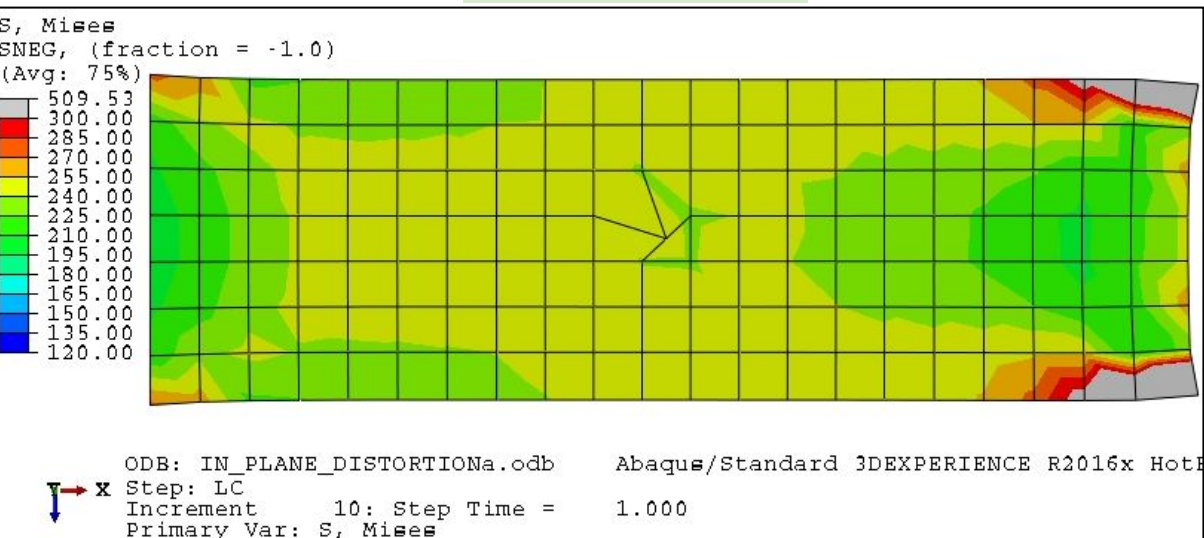
Ideal elements / Reference

Duplicate elements

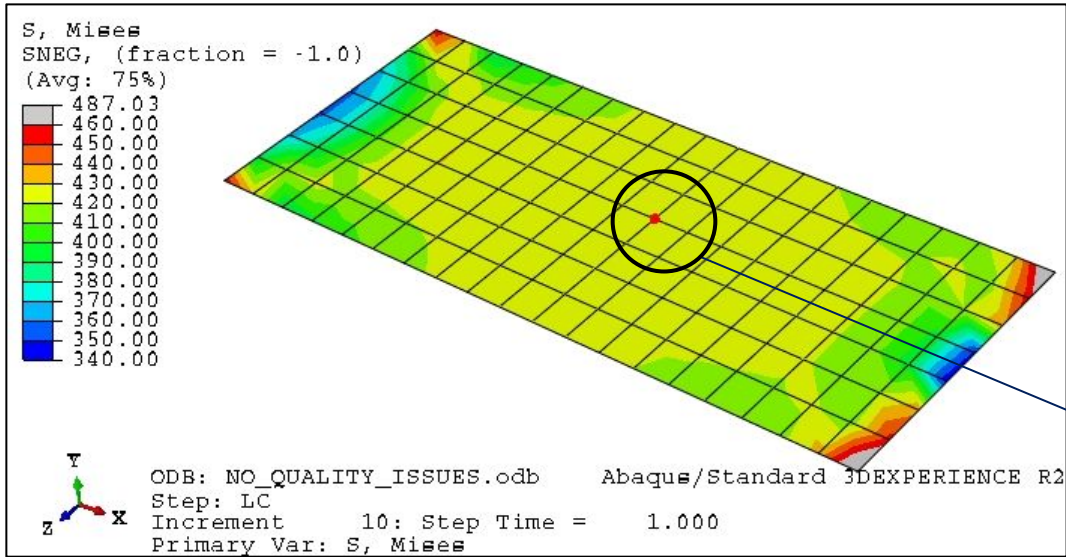


In-plane distortion

Out of plane distortion

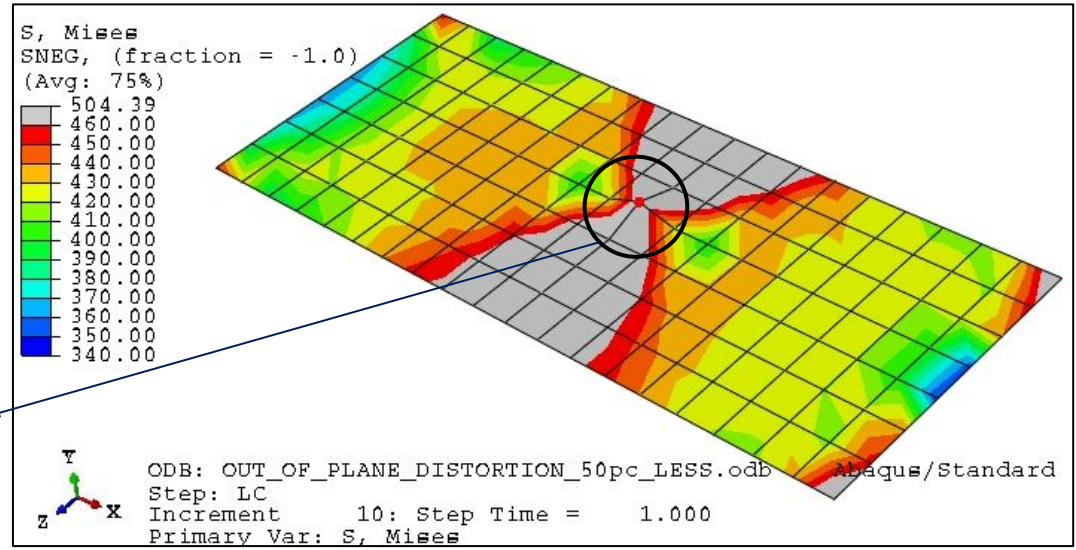


Ideal elements / Reference

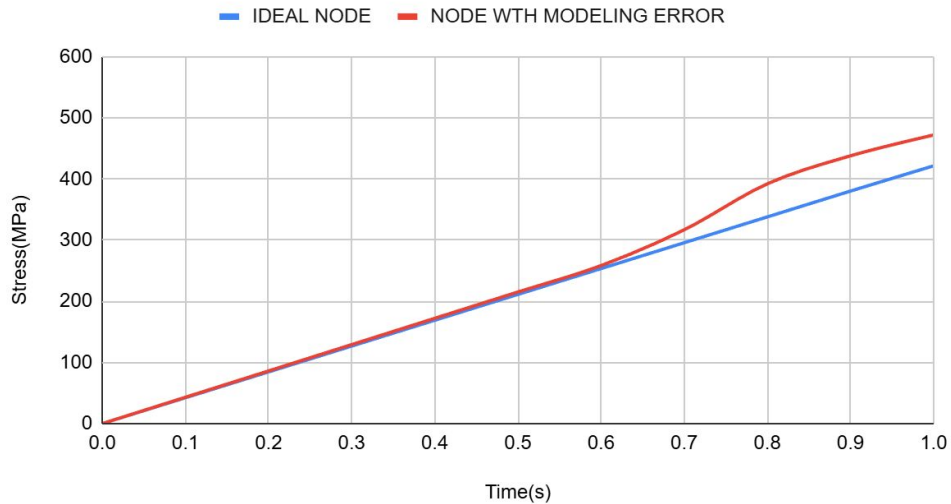


NODE ID:
162205

Out of plane distortion

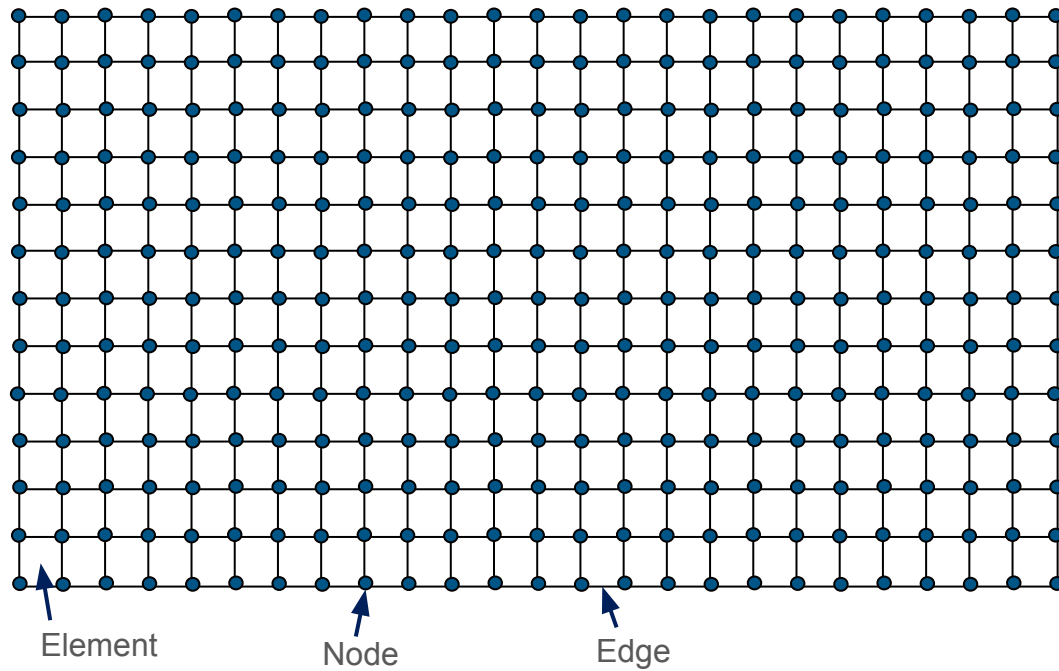


Stress Vs Time @ NODE ID 162205



Graph Extraction from Reference Mesh

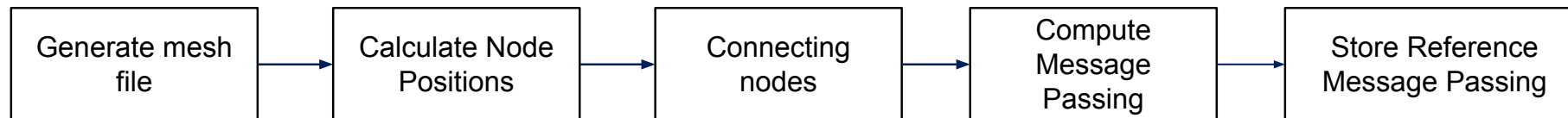
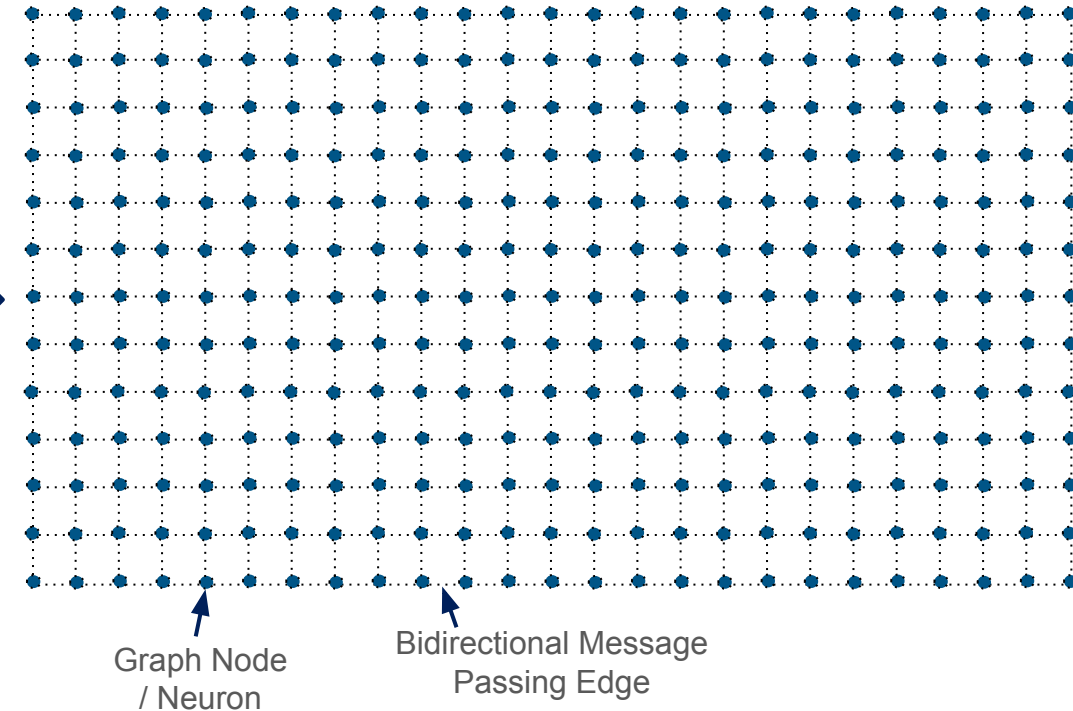
Finite Element Model (FEM)



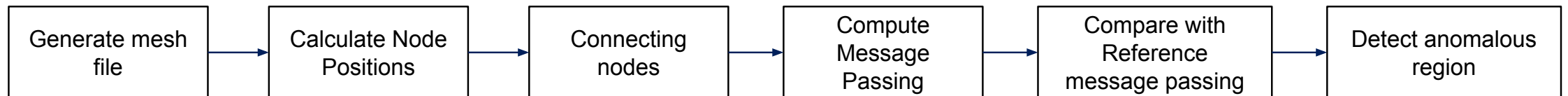
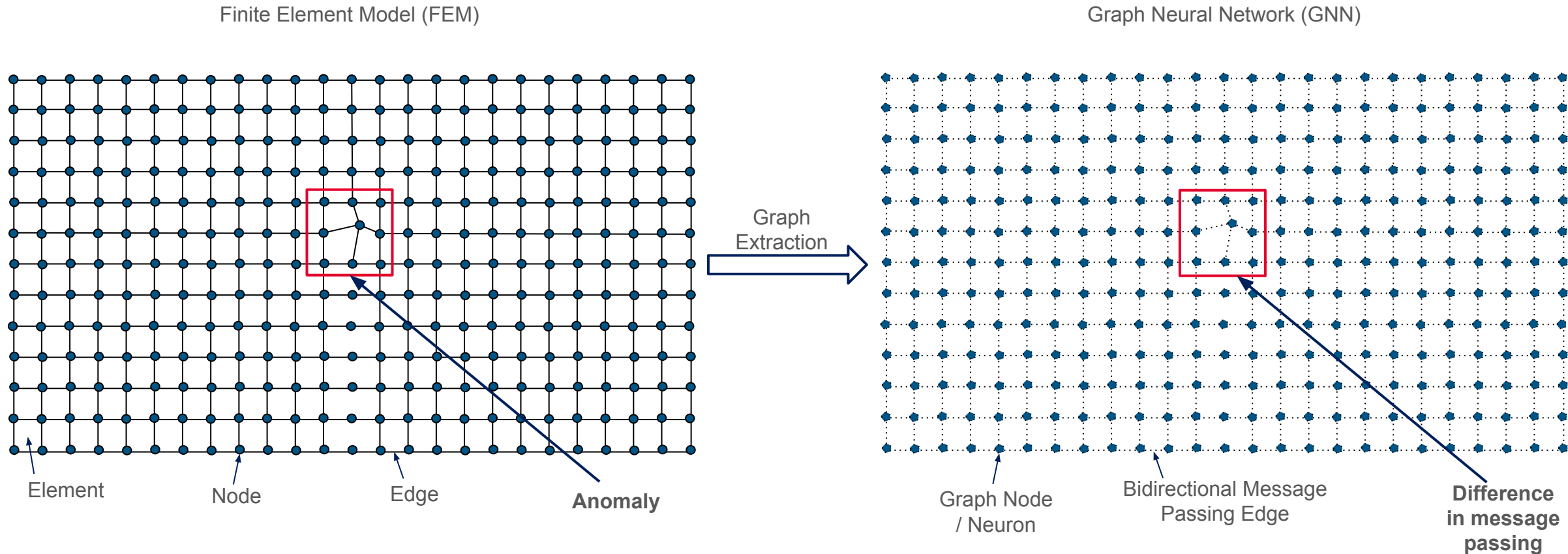
Graph
Extraction

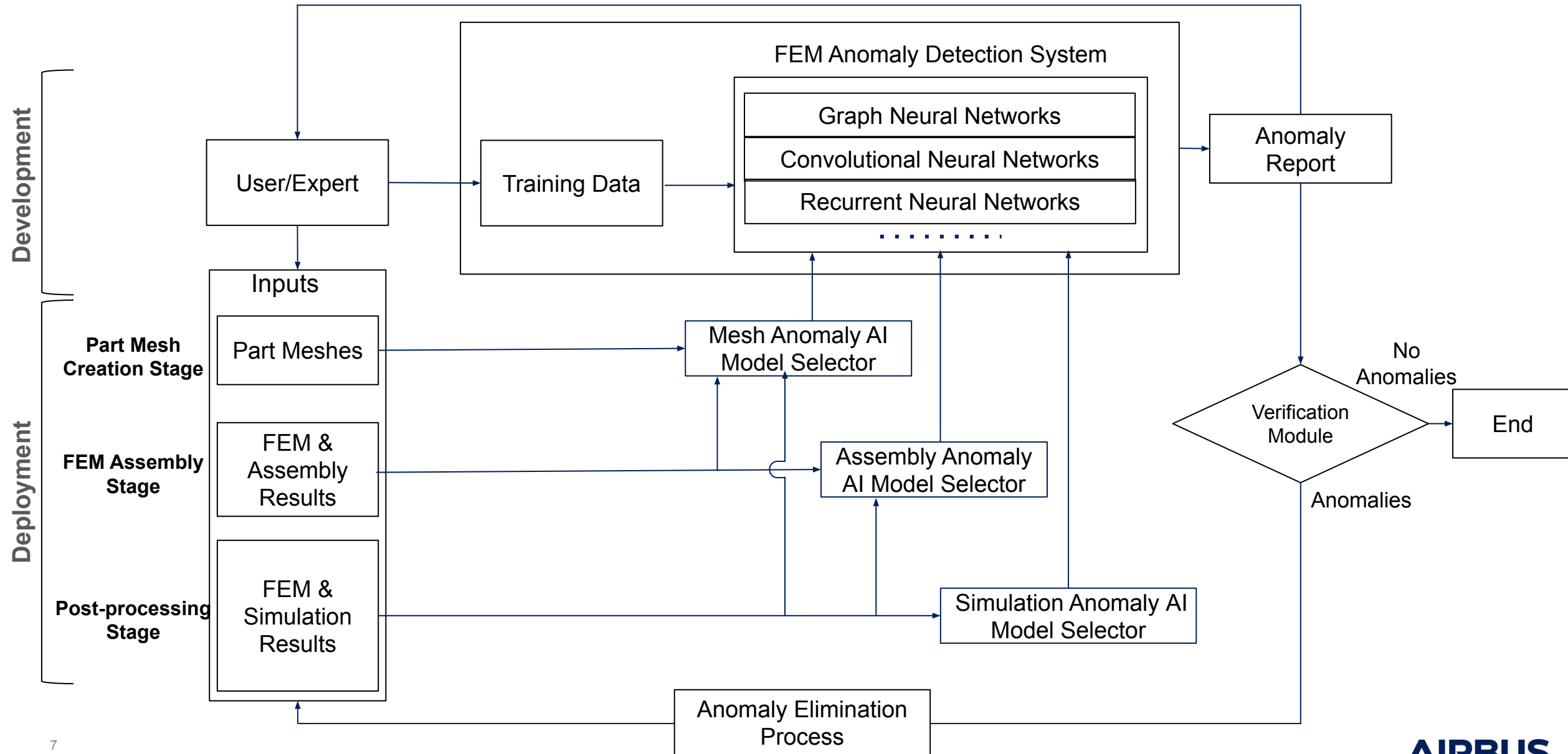


Graph Neural Network (GNN)

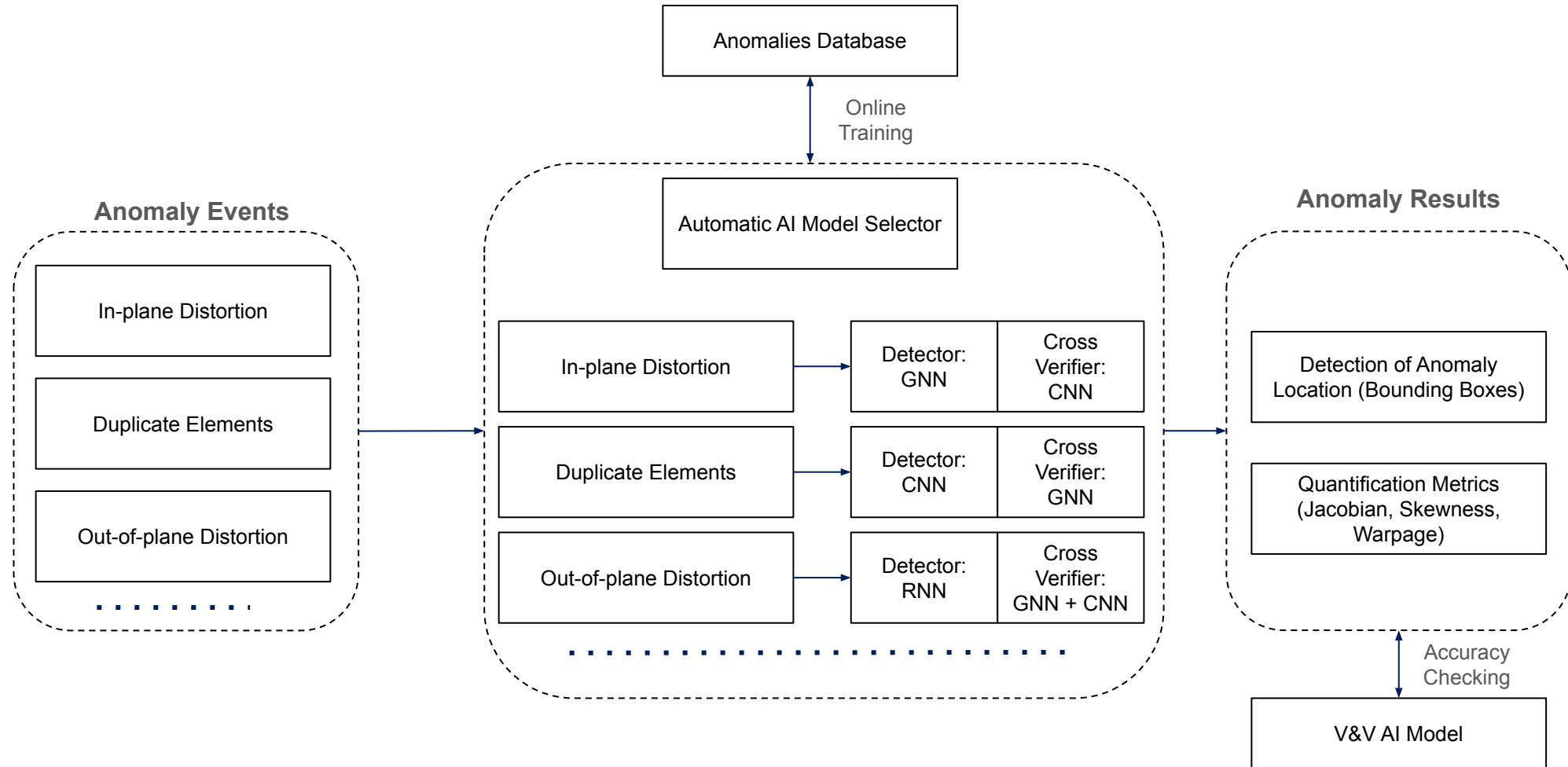


Anomaly Detection Process : In-Plane Distortion

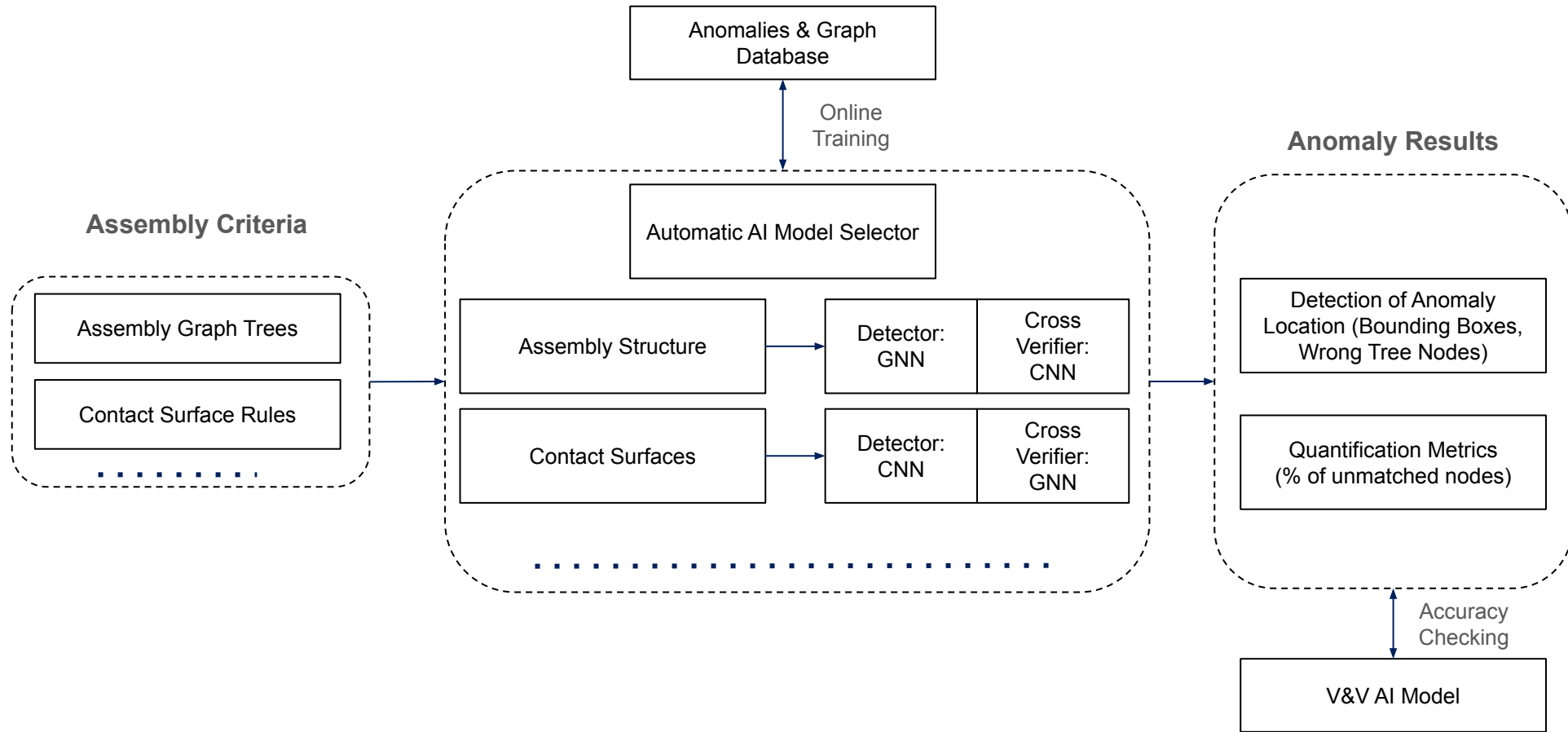




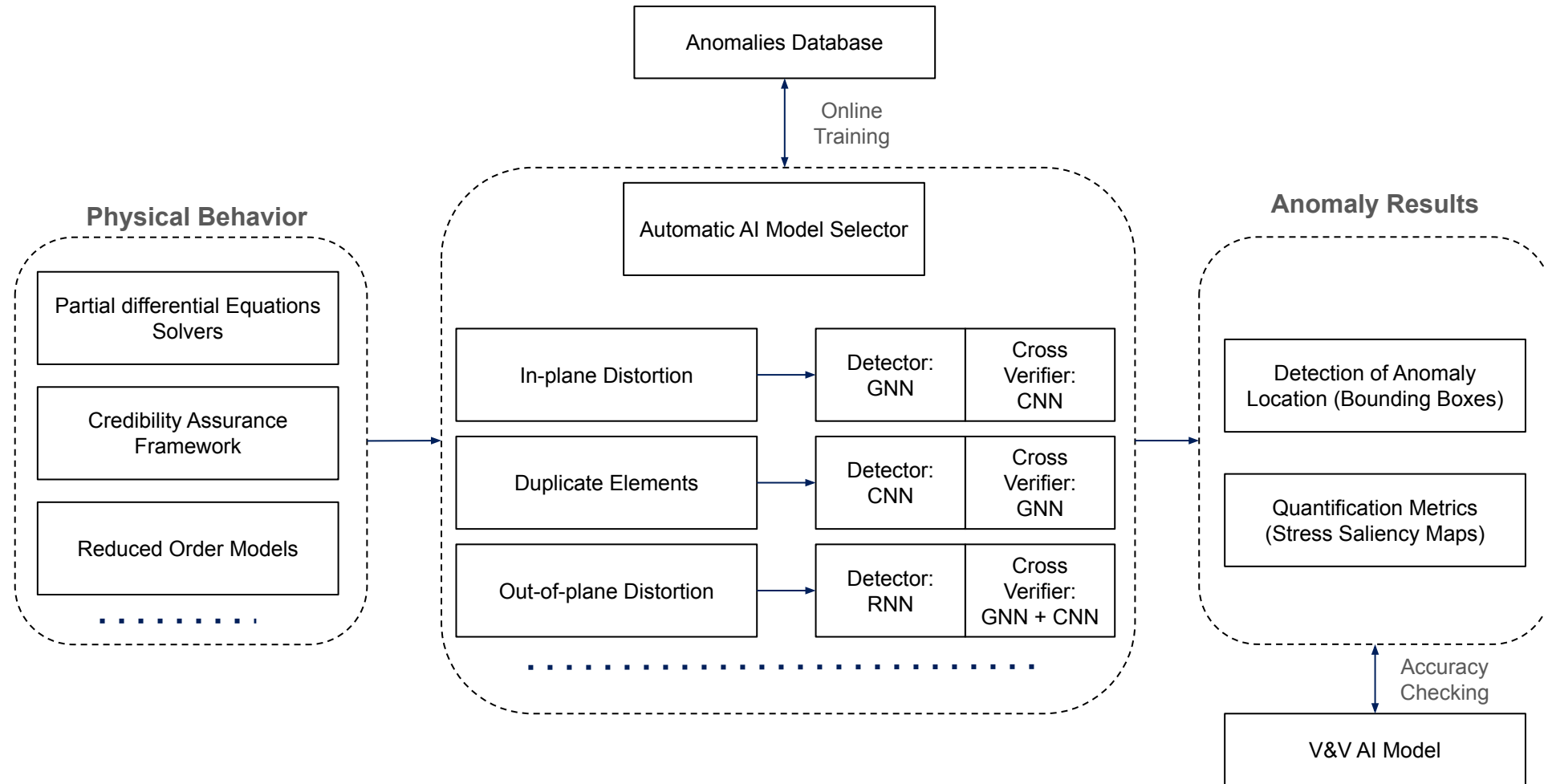
Mesh Anomaly Checker - An example



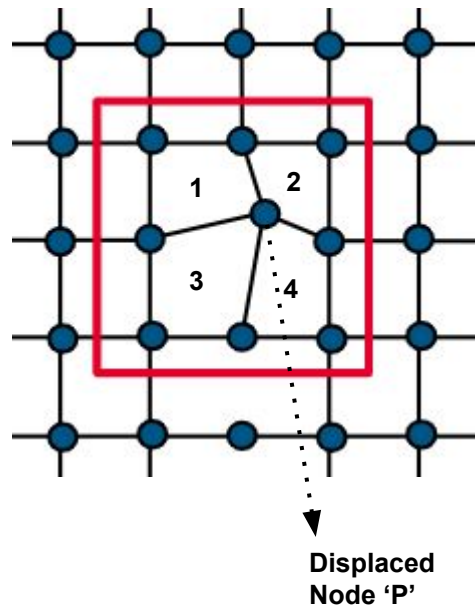
Assembly Anomaly Checker - An example



Simulation Credibility Checker - An example



Anomaly Metric - Jacobian - Quantification



Jacobian Calculation by AI Model

| **Element 1** | Stretched / Convex | **1.2** | The area has increased because **P** moved away from the top-left boundaries. The mapping is stable but the "density" is lower. |

| **Element 2** | Compressed / Convex | **0.5** | **P** moved toward the right boundary, "squeezing" the element. It remains a valid quad, but the Jacobian is smaller. |

| **Element 3** | **Highly Distorted** | **0.1** | The movement of **P** has created a "concave" or "collapsed" shape. At the integration points near the sharpest inward fold, the value drops toward zero or flips negative. |

| **Element 4** | **Inverted / Critical** | **-0.5** | This appears to be the most "folded" element. The nodal ordering relative to the area has likely reversed, leading to a consistently negative Jacobian. |

Thank you

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